CENG381 Stochastic Processes

Chapman-Kolmogorov Equations and Diagonalization of P - Question Set 1

Notation used in this question set:

P^n(i, j)	the n-step transition probability: the probability of being in state j after exactly n steps, starting from state i
C-K equation:	P^(n+m)(i, j) = sum over all intermediate states k of P^n(i, k) * P^m(k, j). In matrix form: P^(n+m) = P^n * P^m
Eigenvalue lam of P:	a scalar such that P*v = lam*v for some non-zero vector v (called the corresponding eigenvector)
Diagonalization:	if P has eigenvalues lam1, lam2 and corresponding eigenvectors, it can be written as P = S_inv * D * S, where D = diag(lam1, lam2) is the diagonal matrix of eigenvalues and S_inv, S are change-of-basis matrices. Then P^n = S_inv * D^n * S, and D^n = diag(lam1^n, lam2^n).

Q1 (30 points). Consider the Markov chain on states {E, W} (East and West lily pads) with transition matrix:

$$\begin{array}{c c} & \begin{array}{cc} E & W \end{array} \\ \begin{array}{c} E \\ W \end{array} & \left[\begin{array}{cc} 0.7 & 0.3 \\ 0.4 & 0.6 \end{array}\right] \end{array}$$

Here p = 0.3 is the probability of jumping from E to W, and q = 0.4 is the probability of jumping from W to E.

a) State the Chapman-Kolmogorov equation for P^(n+m)(E, E) and use it with n = m = 1 to compute P^2(E, E) directly from P.

b) The two eigenvalues of P are lam1 = 1 and lam2 = 1 - p - q. Compute lam2 for p = 0.3, q = 0.4. Verify that lam2 is indeed an eigenvalue by checking det(P - lam2 * I) = 0.

c) Use the diagonalization formula to write down P^n(E, E) explicitly:

P^n(E, E) = q/(p+q) + lam2^n * p/(p+q)

Verify this matches P^2(E, E) from part (a). What does P^n(E, E) approach as n -> infinity?

Q2 (35 points). A 3-state Markov chain models an air conditioner that is either Off (state 0), running at Low power (state 1), or running at High power (state 2). The transition rate matrix Q is given by:

$$\begin{array}{c c} & \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ 0 & \left[\begin{array}{ccc} -1/3 & 1/4 & 1/12 \end{array}\right] \end{array}$$

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1  [  1/3  -1/2    1/6 ]
2  [  1/6   1/3  -1/2  ]
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Q is a RATE matrix (not a probability matrix). Each off-diagonal entry Q(i,j) >= 0 is the rate of jumping from state i to state j. Each diagonal entry Q(i,i) = -(sum of off-diagonal entries in row i) so that each row of Q sums to 0.

a) Verify that each row of Q sums to 0 (check rows 0, 1, and 2).

b) The embedded (jump) chain has transition matrix P where P(i, j) = Q(i, j) / |Q(i, i)| for i != j (i.e., divide each off-diagonal entry by the total rate out of state i). Compute P for this chain.

c) Using the Chapman-Kolmogorov equation P^2 = P * P, compute P^2(0, 2), the probability of being in state 2 after 2 jumps of the embedded chain, starting from state 0.

Q3 (35 points). A Markov chain on the infinite state space {0, 1, 2, 3, ...} has transition probabilities:

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P(i, 0)   = 1 / (i + 1)      for all i >= 0
P(i, i+1) = i / (i + 1)      for all i >= 0
P(i, j)   = 0                for all other j
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a) Verify that each row sums to 1: show that P(i, 0) + P(i, i+1) = 1 for all i >= 0.

b) Apply the Chapman-Kolmogorov equation to compute P^2(0, 0) = sum over all intermediate states k of P(0, k) * P(k, 0). Note that from state 0, the chain can only go to state 0 (with prob 1) or state 1 (with prob 0) -- re-examine P(0, .): since i=0, P(0, 0) = 1/(0+1) = 1. So state 0 is absorbing? Re-read the problem and discuss.

c) Now consider a modified chain where P(0, 1) = 1 (state 0 always moves to state 1). For state i >= 1, define P(i, 0) = 1/(i+1) and P(i, i+1) = i/(i+1). Is state 0 recurrent or transient? Compute the probability that the chain started at state 1 ever returns to state 0 by summing 1/(1*2) + 1/(2*3) + ... and using the telescoping identity 1/(k*(k+1)) = 1/k - 1/(k+1).